FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION:

1. INTRODUCTION

Project Title : UCab – Smart Cab Booking System

Team Members:

Name	Role
Kuruba Sai Mahathi	Frontend Development, Backend Integration, Documentation
Mahesh Patta	Backend Development
Mahi Vedala	Database Design & Testing
Makam VamsiKumar	Documentation
Manaswi Boligarla	Datadase Design

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2. PROJECT OVERVIEW Purpose:

UCab is a web-based cab booking system developed for students and staff to simplify transportation management.

The system allows users to:

- Register and login
- Book rides online
- Calculate fare automatically
- View booking history
- Contact administrators

The main goal is to provide a convenient and efficient transportation booking platform.

Features:

1.User Features

User Registration

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- User Login
- Cab Booking
- Fare Calculation
- Booking History
- Contact Form

2.Admin Features

- Manage booking data
- View user information
- Monitor system activities

3.Additional Features

- Google Maps API Integration
- Distance Calculation
- Travel Time Estimation
- Responsive User Interface

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3. ARCHITECTURE

Frontend Architecture

Frontend is developed using React.js.

Major Components:

- Home Page
- Login Page
- Register Page
- Booking Page
- Booking History Page
- Contact Page
- Navbar
- Footer

Vehicle Cards

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React Router is used for navigation between pages.

Axios is used for API communication.

Backend Architecture

Backend is developed using:

- Node.js
- Express.js Backend follows MVC Architecture.

Flow:

Client
↓
Routes
↓
Controllers
↓
Models
↓
MongoDB

Database Architecture

MongoDB is used as the database.

Collections:

Users

Stores:

- Name
- Email
- Password

Bookings

Stores:

Pickup

-

- Destination
- Journey Date
- Journey Time
- Fare
- Vehicle
- Driver Details **Contacts**

Stores:

- Name
- Email
- Message

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4. SETUP INSTRUCTIONS

Prerequisites

Install:

- Node.js
- MongoDB
- VS Code
- Git

Installation

Clone Repository git

clone <github-link>

Frontend Setup

cd Client npm
install npm run
dev

Backend Setup cd Server
npm install npx nodemon
server.js

Environment Variables

Create .env

PORT=5000

MONGO_URI=mongodb://localhost:27017/cabbooking

JWT_SECRET=mysecretkey

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5. FOLDER STRUCTURE

Client

```
Client
|
├── public
|
├── src
|    ├── components
|    |    ├── Navbar.jsx
|    |    ├── Footer.jsx
|    |    ├── GoogleMap.jsx
|    |    ├── VehicleCard.jsx
|    |    ├── Button.jsx
|    |

|    |
|    |
|    ├── pages
|    |    ├── Home.jsx
|    |    ├── Login.jsx
|    |    ├── Register.jsx
|    |    ├── Booking.jsx
|    |    ├── BookingHistory.jsx
|    |    ├── Contact.jsx
```

```
|   |   └── About.jsx
|   |
|    ├── styles
|    ├── App.jsx
|   └── main.jsx
|
├── package.json
└── vite.config.jsComponents
```

Components

File	Purpose
Navbar.jsx	Navigation bar for all pages
Footer.jsx	Displays footer information
GoogleMap.jsx	Integrates Google Maps API for location selection
VehicleCard.jsx	Displays available vehicle options
Button.jsx	Reusable button component

Pages

File	Purpose
Home.jsx	Landing page of the application
Login.jsx	User login page
Register.jsx	User registration page
Booking.jsx	Main cab booking page
BookingHistory.jsx	Displays previous bookings
Contact.jsx	Contact/feedback page
About.jsx	Information about the project

Server

```
Server
|
├── config
|   └── db.js
|
├── controllers
|    ├── authController.js
|    ├── bookingController.js
|    ├── contactController.js
|
├── models
|    ├── User.js
|    ├── Booking.js
|   └── Contact.js
|
├── routes
|    ├── authRoutes.js
|    ├── bookingRoutes.js
|    ├── contactRoutes.js
|
├── .env
├── package.json
└── server.jsConfig
```

Config : Database connection.

File	Purpose
db.js	Connects the application to MongoDB

Controllers: Business Logic.

File	Purpose
authController.js	Handles user registration and login
bookingController.js	Creates, retrieves, updates, and deletes bookings
contactController.js	Stores user contact messages

Models: MongoDB Schemas.

File	Purpose
User.js	Defines the schema for user information
Booking.js	Defines the schema for booking details
Contact.js	Defines the schema for contact messages

Routes: API Endpoints.

File	Purpose
authRoutes.js	Authentication API endpoints
bookingRoutes.js	Booking-related API endpoints
contactRoutes.js	Contact form API endpoints

Main Backend Files

File	Purpose
server.js	Entry point of the backend, configures Express, middleware, and routes
.env	Stores

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6. RUNNING THE APPLICATION Frontend

cd Client npm run dev

Runs frontend at:

http://localhost:5173

Backend

cd Server

npx nodemon server.js Runs

backend at:

http://localhost:5000

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7. API DOCUMENTATION

Authentication APIs

Register

POST /api/auth/register

Login

POST /api/auth/login

Booking APIs

Create Booking

POST /api/bookings

Get Booking History

GET /api/bookings/:userId

Update Booking

PUT /api/bookings/:id

Delete Booking

DELETE /api/bookings/:id

Contact APIs :**Submit Contact Form**

POST /api/contact

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8. AUTHENTICATION

Authentication is implemented using:

- JWT (JSON Web Token)
- Password Encryption using bcryptjs

Process

1. User registers.
2. Password is encrypted.
3. User logs in.
4. JWT token is generated.
5. User accesses protected routes.

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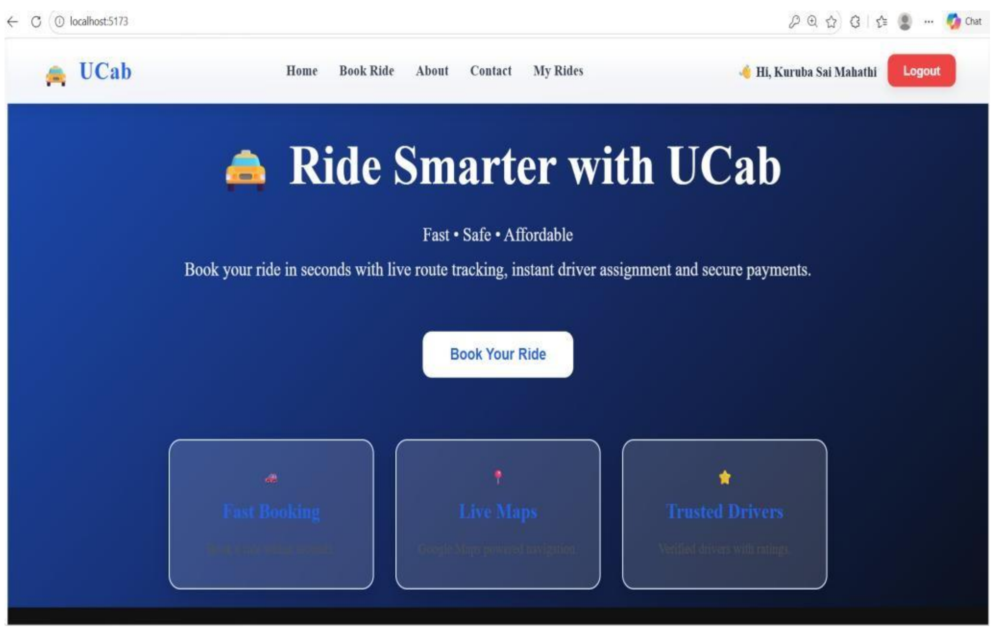
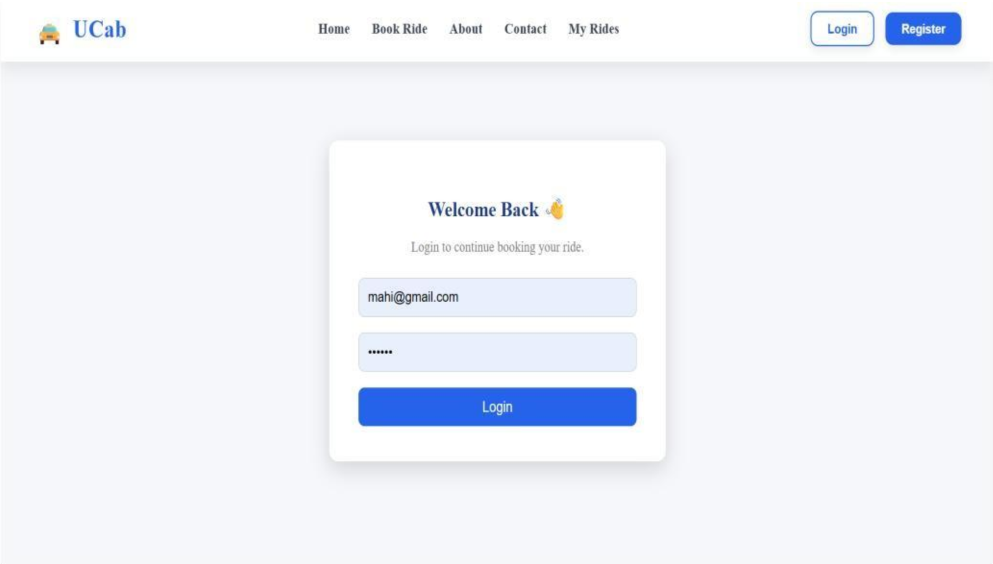
9. USER INTERFACE

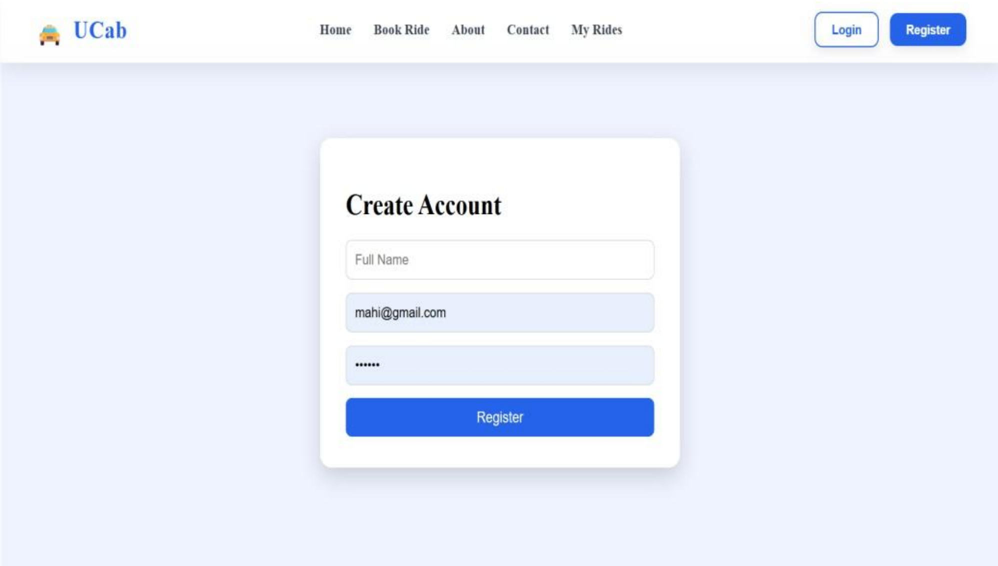
Home Page:



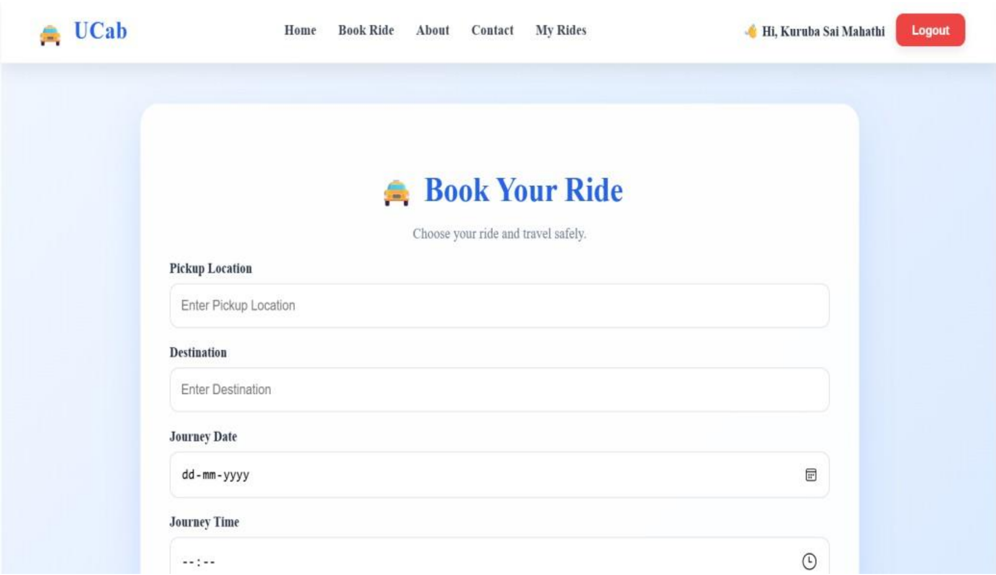
Login Page:

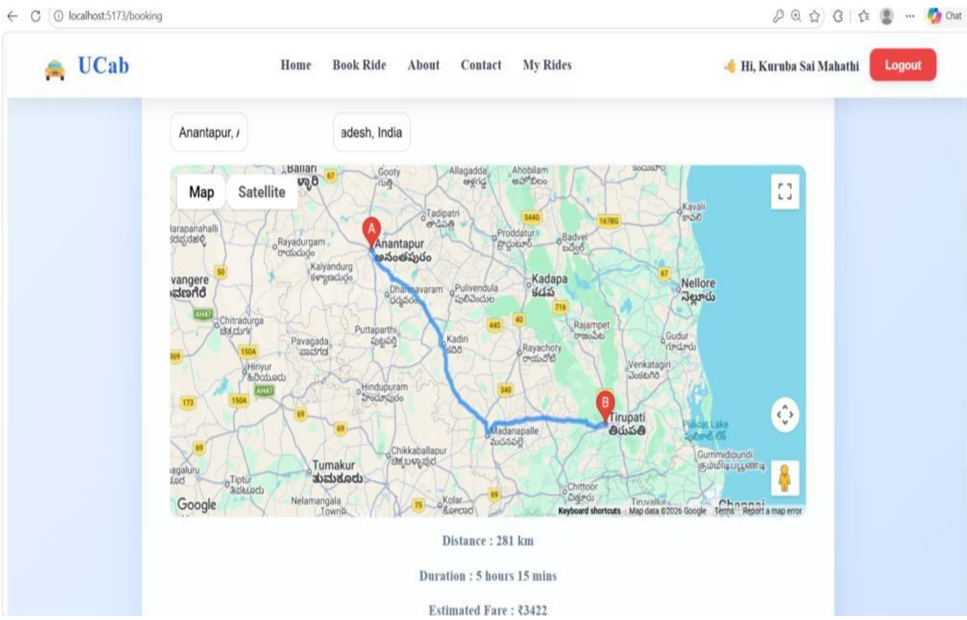


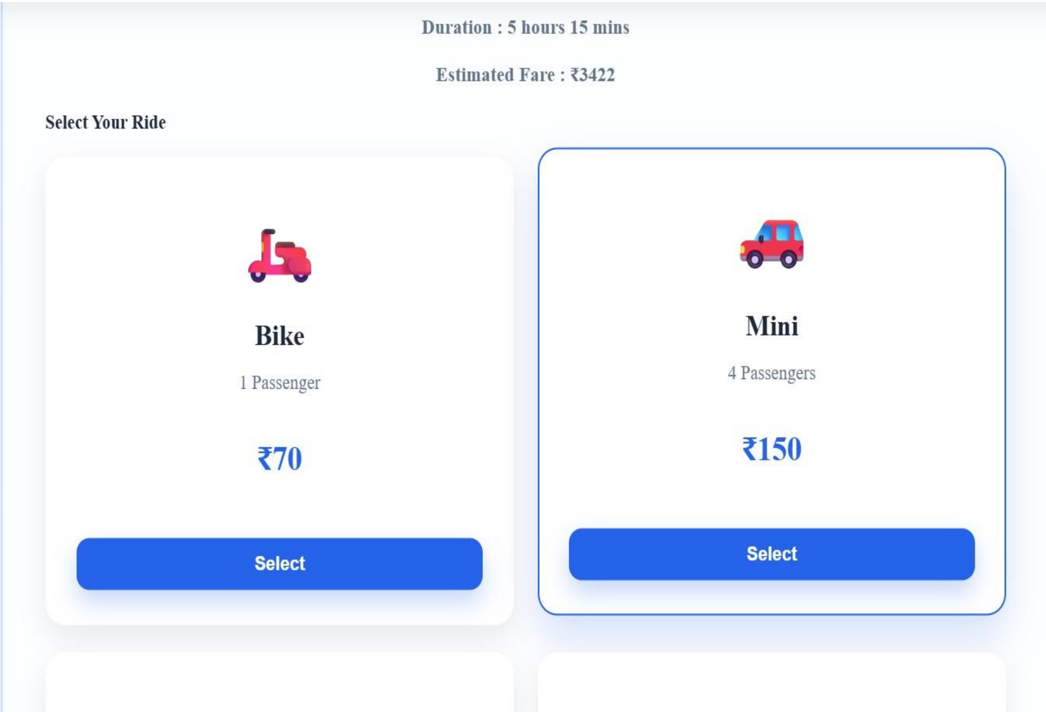
Register Page:

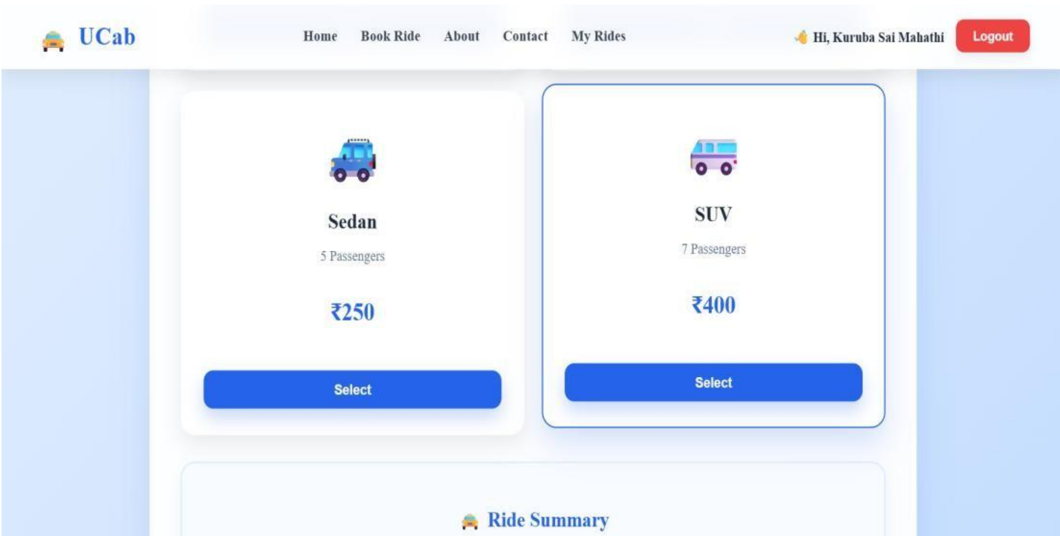


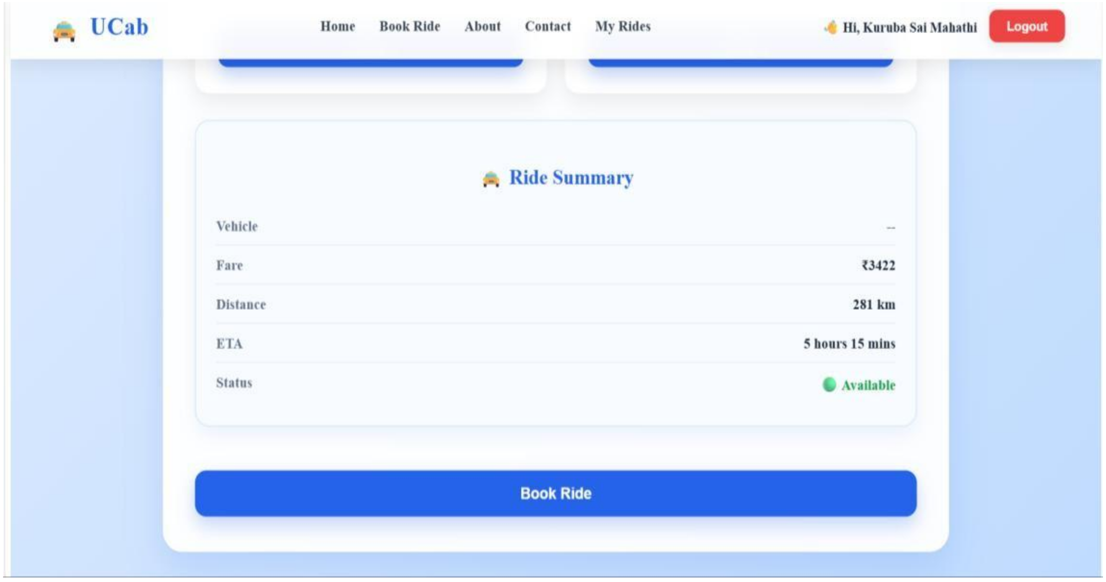
Booking Page:


UCab
Home
Book Ride
About
Contact
My Rides
Hi, Kuruba Sai Mahathi
Logout
Book Your Ride
Choose your ride and travel safely.
Pickup Location
Enter Pickup Location
Destination
Enter Destination
Journey Date
dd-mm-yyyy
Journey Time
--:--

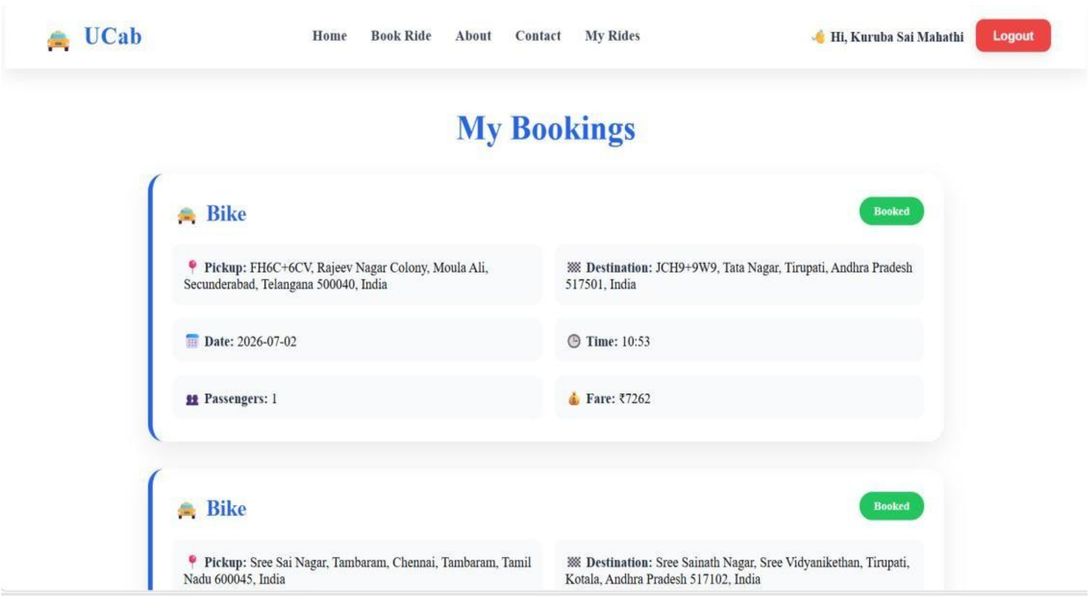

localhost:5173/booking
Chat
UCab
Home
Book Ride
About
Contact
My Rides
Hi, Kuruba Sai Mahathi
Logout
Anantapur, /
adesh, India
Map
Satellite
Anantapur
Kadapa
Nellore
Tirupati
Madanapalle
Tumakur
Google
Keyboard shortcuts
Map data ©2026 Google
Terms
Report a map error
Distance : 281 km
Duration : 5 hours 15 mins
Estimated Fare : ₹3422


Duration : 5 hours 15 mins
Estimated Fare : ₹3422
Select Your Ride
Bike
1 Passenger
₹70
Select
Mini
4 Passengers
₹150
Select

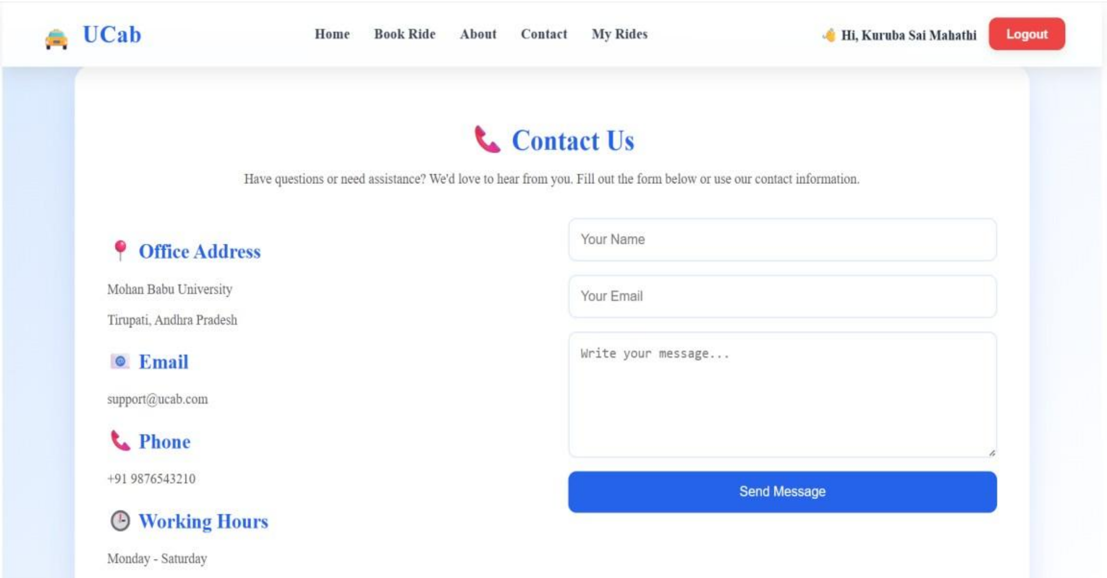

UCab
Home
Book Ride
About
Contact
My Rides
Hi, Kuruba Sai Mahathi
Logout
Sedan
5 Passengers
₹250
Select
SUV
7 Passengers
₹400
Select
Ride Summary



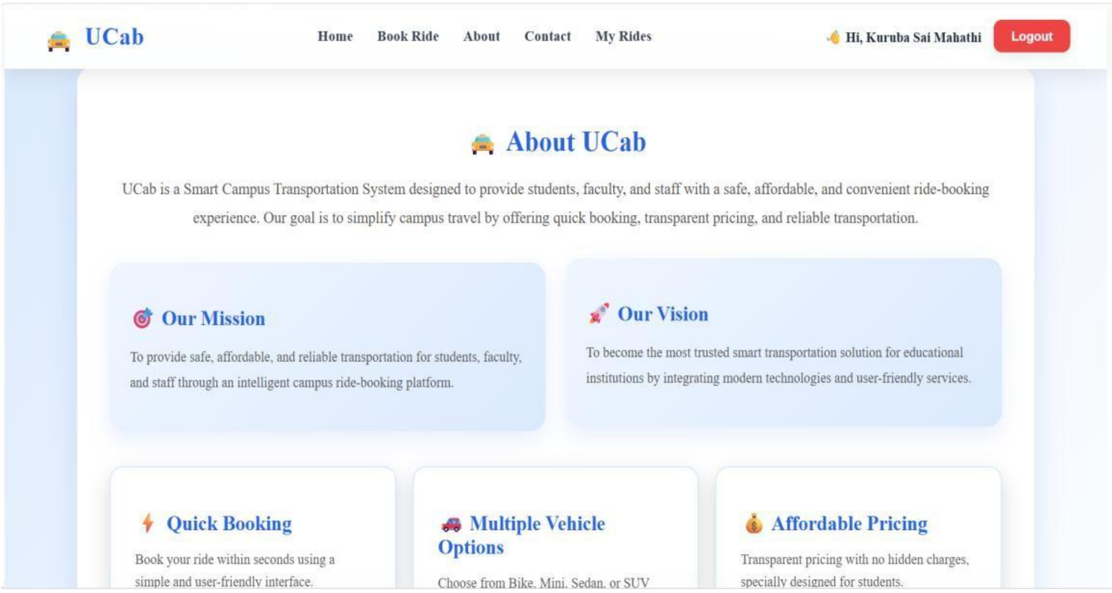
Booking History Page:



Contact Page:



About Page:



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10. TESTING

Testing Performed:

Functional Testing

- User Registration
- Login

- Booking Creation
- Booking History
- Contact Submission

Database Testing

- Data Storage Verification
- CRUD Operations Validation

UI Testing

- Navigation
- Form Validation
- Responsive Design

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11. SCREENSHOTS OR DEMO

GitHub Repository

https://github.com/Mahathikuruba/CabBooking.git

Demo Video

https://drive.google.com/file/d/1GttBMyTOPzp3S_fgu1BLtFxYO_2_vnQr/view?usp=sharing

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12. KNOWN ISSUES

- No online payment integration.
- No live cab tracking.
- Limited admin functionality.
- Works best with stable internet connection.

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13. FUTURE ENHANCEMENTS

Future Improvements:

- Online Payment Gateway
- Live GPS Tracking
- Ride Sharing
- AI-Based Demand Prediction
- Driver Management System

- Email Notifications
- Mobile Application Development

Conclusion:

UCab successfully provides an efficient online cab booking platform using the MERN Stack. The system demonstrates frontend development, backend API integration, database management, authentication, and Google Maps API integration.